# Alexander Adams

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## EDUCATION

**University of Pittsburgh, Pittsburgh, PA** **Expected Graduation 2028**

- BS in Mechanical Engineering
- BS in Applied Mathematics
- Minor in Electrical Engineering

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, C, MATLAB, Java, JavaScript, HTML/CSS
**Tools & Libraries:** NumPy, pandas, scikit-learn, numerical PDEs, optimization, clustering, applied linear algebra

## WORK EXPERIENCE

**Carnegie Mellon University, Undergraduate Researcher** **Aug 2025 – Present**

- Developing 4-layer deep embedded clustering autoencoder (DEC/IDEC) neural networks on vector pair correlation functions (vPCFs) to cluster and identify key structural features in ferroelectric materials.
- Trained DEC/IDEC machine learning models on 1,000+ STEM-derived vPCF embeddings, evaluating cluster quality across 5–20 clusters using unsupervised metrics.

**University of Pittsburgh, Undergraduate Researcher** **Aug 2025 – Present**

- Developed a Bayesian-optimized random forest (BayesSearchCV) predicting bonded thin-film thickness for dip-coated PDMS across three solvent systems, achieving out-of-fold $R^2 = 0.9734$ - the top result across 5 regression families and 3 ensemble strategies tested.
- Diagnosed a severe distribution-shift failure under sequential holdout validation ($R^2$ as low as $-331.96$) and resolved it via stratified 5-fold cross-validation and a bootstrap/Cholesky-correlated-noise pipeline that expanded a 58-sample experimental dataset to 3,000 samples with PubChem-derived solvent descriptors.

**National Science Foundation, Machine Learning REU Intern** **Jun 2026 – Aug 2026**

- Engineered richer node-level (8 features) and graph-level (10+ features) representations for a GCN pipeline predicting Voronoi lattice stiffness, improving test $R^2$ from 0.9148 to 0.9456 and reducing test RMSE from 0.003818 to 0.002882 over the baseline model.
- Extended 5-seed benchmarking to residual-connection and Huber-loss variants alongside 3 optimizers and 4 graph neural network architectures (GCN-2, GCN-3, GraphSAGE, GIN); promoted plain 3-layer GCN as the top mean test-accuracy configuration while a learning-rate retune cut residual-model worst-seed $R^2$ variance (std 0.0115 → 0.0080).
- Quantified model uncertainty via a 5-member ensemble (test $R^2 = 0.9528$) to estimate prediction-interval coverage, measured out-of-sample generalization decay under distribution shift (external $R^2 = 0.4024$), and stress-tested robustness through reflection-invariance checks across mirrored lattice geometries (<3% mean deviation).

**Limex, Quantitative Trading Intern** **Aug 2025 – Sep 2025**

- Developed statistical arbitrage, breadth scalping, and momentum breakout algorithmic trading strategies for the Limex Trading Challenge using Python, pandas, NumPy, and asyncio, executing across 40–50+ liquid equities.
- Engineered multi-asset, low-latency execution engines with async API integration to scan 300+ opportunities per minute reducing execution latency by ~60%, and increasing trade logging/real-time PnL tracking accuracy by 35%.

## PUBLICATIONS

- Comparing the Moore-Penrose Pseudoinverse and Gradient Descent for Solving Linear Regression Problems: A Performance Analysis, arXiv Preprint, May 2025

## PROJECTS & COMPETITIONS

**Hudson River Trading Macro Placement Challenge** **Mar 2026 – May 2026**

- Developed a multi-strategy greedy macro placement algorithm combining 5 sorting heuristics with HPWL-based wirelength estimation across 18 IBM ICCAD04 benchmarks; achieved 100% placement validity (zero overlaps) and outperformed the simulated annealing baseline on 11/18 benchmarks (61%) with 3.9s average runtime.

**IMC Trading Prosperity 4 Trading Challenge** **Apr 2026 – May 2026**

- Backtested statistical arbitrage and momentum strategies across 7 simulated instruments using historical round data; tuned position sizing and entry threshold parameters across 50+ optimization iterations.

**Quantitative Portfolio Optimization** **Jun 2025**

- Engineered a MPT optimizer to allocate capital across 5 equities using historical price data. Constructed the efficient frontier via Monte Carlo simulations and constrained optimization on the covariance matrix, achieving a 44% volatility reduction and 46% Sharpe ratio improvement over naive allocations.